Alg 2 Notes 7.1 – EXPLORING EXPONENTIAL MODELS

I. GENERAL FORM OF EXPONENTIAL EQUATIONS

have a variable in the exponent (power)

① original (initial) amount

$y = a \cdot b^x$ ← time

*Do Not multiply a to b!

amount after time

your base that's raised to the power

$y = a(b)^x$

Principle (original amount)

$A = P(1 \pm r)^t$ ← time of experiment

amount after time

plus or minus

r = rate as a decimal

[why 2 formulas? b used when words are "doubling, halving"]

II. INDENTIFYING IF IT IS EXPONENTIAL GROWTH OR DECAY

Look at b: If $b > 1$ then it is growth If $(1 \pm r) > 1$ then it is growth

If $0 < b < 1$ then it is decay If $0 < (1 \pm r) < 1$, then it is decay

Examples. Circle if the equation is growth or decay.

tricky!

1. $y = 5(3)^x$ $a(b)^x$ — Growth / Decay

2. $y = 2\left(\frac{2}{3}\right)^x$ — Growth / Decay

3. $y = 50(1.7)^x$ — Growth / Decay

4. $y = 4(0.8)^x$ — Growth / Decay

5. $y = 3(4)^{-x}$ — Growth / Decay

$y = 3(4)^{-1 \cdot x}$

$= 3(4^{-1})^x$

$= 3\left(\frac{1}{4}\right)^x$

III. HOW TO FIND THE PERCENTAGE RATE

Example: Given the following exponential function, identify whether the change represents growth or decay and determine the percentage rate of increase of decrease.

6. $y = 50(1.7)^x$

"therefore"

$b > 1 \therefore$ growth (increase)

$b = 1 + r$

$r = .7$ rate as a decimal

"% means out of 100"

To stop or a % sign and be equivalent, we must multiply by 100

70 Move decimal 2 places to the right (mult by 100)

$r = 70\%$ = percentage rate

7. $y = 4(0.8)^x$

Decay

$b = 1 - r$

To solve for r, replace b.

$.8 = 1 - r$

$-.2 = -r$

$r = .2$

Decay

$= .2(100)\%$

20%

DECAY

or

Decrease

III. INTERPRET EXPONENTIAL [redacted] CONSTANT

Examples:

exponential constant

The function $f(t) = 7500(1.025)^t$ represents the change in a quantity over t minutes. What does the constant 1.025 reveal about the rate of change of the quantity?

$1 + r = 1.025$

$r = .025 \rightarrow 2.5\%$

The function is growing exponentially at a rate of 2.5 % every minute

*

The function $f(t) = 4600(0.9)^{\frac{t}{24}}$ represents the change in a quantity over t days. What does the constant 0.9 reveal about the rate of change of the quantity?

$t = 1$ day

Replace t with $\frac{t}{24}$

$\frac{t}{24}$ = a day $\rightarrow$ $t = 24$ days

$1 - r = .9$

$-r = -.1$

$r = .1 \rightarrow \frac{10}{100} = 10\%$

The function is decaying exponentially at a rate of 10 % every * 24 days.

The function $f(t) = 5300(0.35)^{365t}$ represents the change in a quantity over t years. What does the constant 0.35 reveal about the rate of change of the quantity?

Decay $\rightarrow$ $.35 = 1 - r$

$-.65 = -r$

$r = .65$

$\frac{365 \text{ days}}{1 \text{ year}}$

$t = 1$ year

$365t = 1$ year

$t = \frac{1}{365}$ years $= 1$ day

The function is decaying exponentially at a rate of 65 % every $\frac{1}{365}$ years or 1 day

A2 Practice 7.1 Name:______________________ Per:_____

1. Given the following exponential function, identify whether the change represents growth or decay, and determine the percentage rate of increase or decrease.

$$y = 4700(0.91)^x$$

$.91 < 1$ so

$.91 = \boxed{91\% \quad \text{decay}}$

2. Given the following exponential function, identify whether the change represents growth or decay, and determine the percentage rate of increase or decrease.

$$y = 8000(1.03)^x$$

$.03$ $\boxed{\text{growth}}$

$\boxed{3\%}$

3. The function $f(t) = 980(0.45)^{10t}$ represents the change in a quantity over t decades. What does the constant 0.45 reveal about the rate of change of the quantity?

The function is decaying (word bank 1) exponentially at a rate of 55 % every $\frac{1}{10}$ decade (word bank 2) or 1 year.

Word bank 1: (a) decaying, (b) growing

Word bank 2: (a) 10 months, (b) decade, (c) 10 decades, (d) month, (e) year

$.45 = 1 - r$

$-.55 = -r$

$r = .55 \rightarrow 55\%$

$t = 1$ decade

$10t = 1$ decade

$t = \frac{1}{10}$ ~~decay~~ decade

or

1 year

4. The function $f(t) = 3800(0.935)^{365t}$ represents the change in a quantity over t years. What does the constant 0.935 reveal about the rate of change of the quantity?

The function is decaying (word bank 1) exponentially at a rate of 6.5 % every $\frac{1}{365}$ year (word bank 2) or 1 day

Word bank 1: (a) decaying, (b) growing

Word bank 2: (a) day, (b) 365 years, (c) 365 weeks, (d) year, (e) week

$.935 < 1$ so decay

$1 - r = .935$

$-r = -.065$

$r = .065$

6.5%

$t = 1$ year

$365t = 1$ year

$t = \frac{1}{365}$

5. A town has a population of 10000 and grows at 2.5% every year. What will be the population after 7 years, to the nearest whole number?

$A(t) = P(1 \pm r)^t$

rate as a decimal

$r = .025$

$A(7) = 10{,}000(1 + .025)^7$

$= 11886.85$

$= \boxed{11{,}887 \text{ pop.}}$

6. An element with mass 140 grams decays by 21.6% per minute. How much of the element is remaining after 20 minutes, *to the nearest 10th of a gram*?

$.216$

$A(20) = 140(1 - .216)^{20}$

$\boxed{= 1.1\text{g}}$